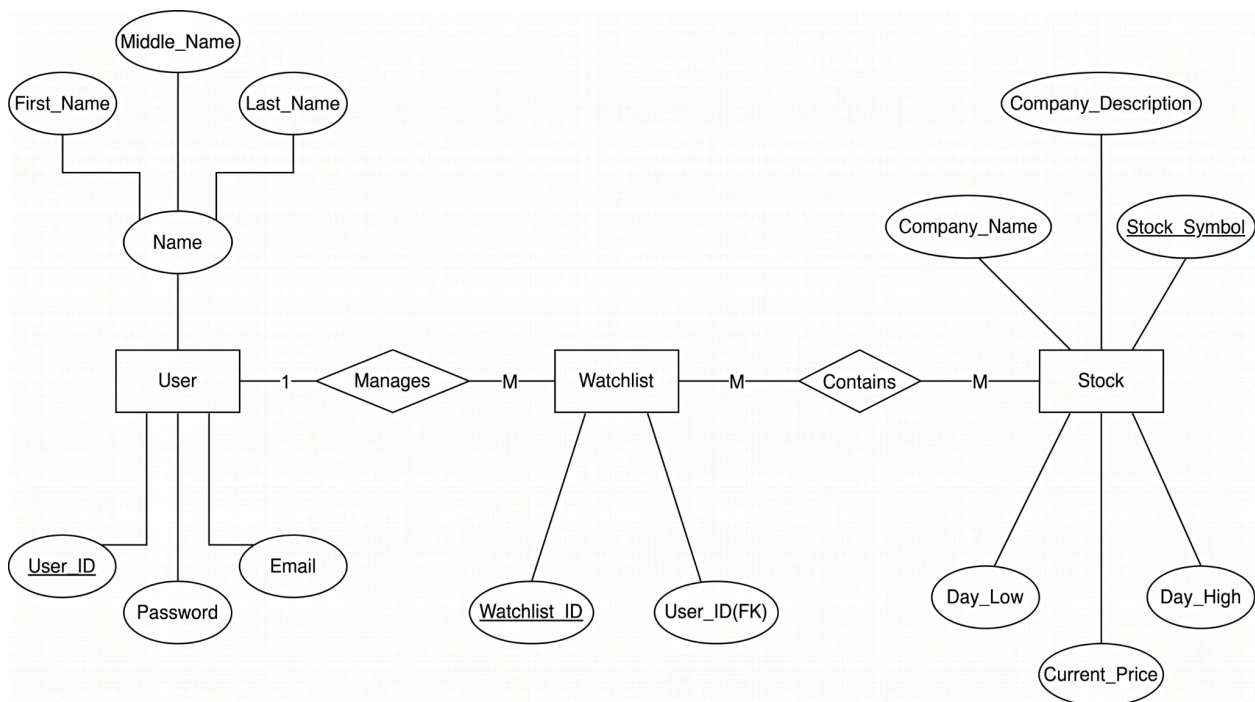


Lab - 4

Aim: To design and implement the MongoDB database for the selected project by creating collections, inserting sample data, and performing CRUD operations.

Part - A - ER Diagram :



Part B – MongoDB Database Creation

MongoDB Compass - Frenil/Shell

Connections Edit View Help

Compass

() My Queries

⚙ Data Modeling

CONNECTIONS ()

Search connections

▼ Frenil

- admin
- config
- equitrackDB
 - users
 - watchlists

mongosh Frenil watchlists

```
> use("config")
> switched to db config
> use equitrackDB
> switched to db equitrackDB
> db.createCollection("users")
> db.createCollection("watchlists")
{ ok: 1 }
> db.users.insertOne({
  fullName: "Frenil Vaghasta",
  email: "24ceus162@ddu.ac.in",
  password: "Frenil@393",
  createdAt: new Date()
})
{
  acknowledged: true,
  insertedId: ObjectId('6a81b7cb7c7a1e23c3e1b78')
}
> db.watchlists.insertOne({
  userEmail: "24ceus162@ddu.ac.in",
  companyName: "TechCorp",
  stockSymbol: "TCORP",
  addedAt: new Date()
})
{
  acknowledged: true,
  insertedId: ObjectId('6a81b88ab7c7a1e23c3e1b79')
}
> db.users.findOne({ email: "24ceus162@ddu.ac.in" })
{
  _id: ObjectId('6a81b7cb7c7a1e23c3e1b78'),
  fullName: 'Frenil Vaghasta',
  email: '24ceus162@ddu.ac.in',
  password: 'Frenil@393',
  createdAt: 2026-08-16T13:17:49.904Z
}
> db.watchlists.find({ userEmail: "24ceus162@ddu.ac.in" })
{
  _id: ObjectId('6a81b88ab7c7a1e23c3e1b79'),
```

MongoDB Compass - Frenil/Shell

Connections Edit View Help

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  password: 'Frenil@393',
  createdAt: 2026-08-16T13:17:49.904Z
}
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{
  _id: ObjectId('6a81b88ab7c7a1e23c3e1b79'),
  userEmail: '24ceus162@ddu.ac.in',
  companyName: 'TechCorp',
  stockSymbol: 'TCORP',
  addedAt: 2026-08-16T13:18:02.097Z
}
> db.users.updateOne(
  { email: "24ceus162@ddu.ac.in" },
  { $set: { fullName: "Frenil V." } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.watchlists.deleteOne({
  userEmail: "24ceus162@ddu.ac.in",
  stockSymbol: "TCORP"
})
{
  acknowledged: true,
  deletedCount: 1
}
> db.users.insertOne({
  fullName: "Frenil Vaghasta",
  email: "24ceus162@ddu.ac.in",
  password: "hashed_password_123",
  createdAt: new Date()
})
{
  acknowledged: true,
  insertedId: ObjectId('6a81b8d7b7c7a1e23c3e1b7a')
}
> db.watchlists.insertOne({
  userEmail: "24ceus162@ddu.ac.in",
  companyName: "TechCorp",
  stockSymbol: "TCORP",
```

